Finding the best feasible bundle

Lecture notes

What this lecture is about. Lecture 9 went through the utility functions, one form at a time (see the L09 notes). Today we use them. First derivatives, marginal utility and the MRS. Then four kinds of preferences on one budget. Then the gradient, the utility mountain, the price arrow and the budget roof, drawn one at a time and then together. Then the condition $\nabla u = \lambda \nabla c$, the Lagrangian, and a simpler way with one variable. We end with perfect substitutes. The interactive graphs are on the slides; these notes follow the same order.

1 Derivatives of our functions

A derivative is a slope. For $f(x) = x^2$: $f'(x) = 2x$, and in general x^n has derivative $n\,x^{n-1}$. A partial derivative changes one variable and holds the other fixed. For $u = x\,y$: $\partial u/\partial x = y$ and $\partial u/\partial y = x$. These are the marginal utilities, MU_x and MU_y.

	u	$\dfrac{\partial u}{\partial x}$	$\dfrac{\partial u}{\partial y}$
linear	$ax + by$	a	b
Cobb-Douglas	$x^\alpha y^\beta$	$\alpha x^{\alpha-1} y^\beta$	$\beta x^\alpha y^{\beta-1}$
quasi-linear	$\alpha x + y^\beta$	α	$\beta y^{\beta-1}$

Leontief. The tangency is always at the kink. So find where the kinks, $ax = by$, meet the budget line.

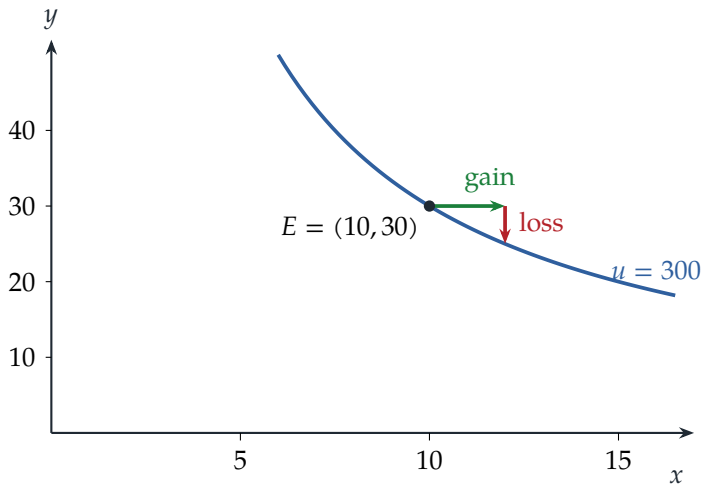
2 The general way: gain = loss

I am on a specific indifference curve and I move along the curve. What does this mean algebraically?

$$\mathrm{MU}_x\,dx + \mathrm{MU}_y\,dy = 0.$$

Or another way of thinking: whatever utility gain (loss) I make by having a tiny bit more x should be equal to whatever utility loss (gain) I make by having a tiny bit less y (assuming monotonicity for both goods). This is the MRS, and it is equal to

$$\mathrm{MRS} = \frac{\mathrm{MU}_x}{\mathrm{MU}_y}.$$

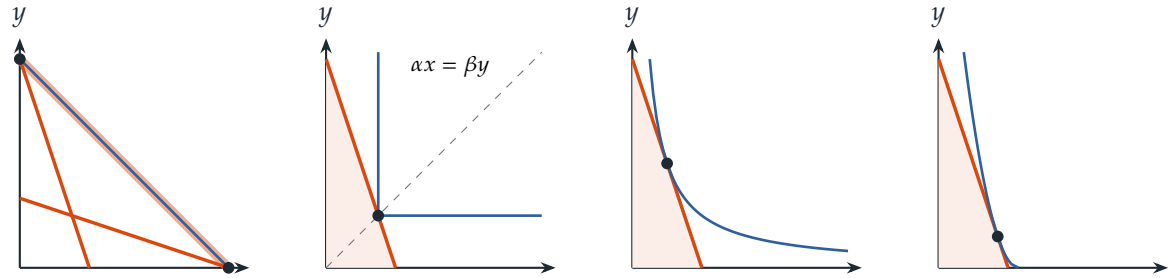


The example, worked in class. For $u = xy$ on the curve $u = 300$ through $E = (10, 30)$: $\text{MU}_x = y$ and $\text{MU}_y = x$, so

$$\text{MRS} = \frac{y}{x} = \frac{30}{10} = 3.$$

3 One budget, four kinds of preferences

The budget for the pictures: prices $p_x = 3$, $p_y = 1$, money $B = 60$, so $3x + y = 60$. In each panel the curves are blue, the budget is orange and the black dot is the best bundle. The pictures show the case $\alpha = \beta = 1$.



linear $u = \alpha x + \beta y$ | Leontief $u = \min\{\alpha x, \beta y\}$ | Cobb-Douglas $u = x^\alpha y^\beta$ | quasi-linear $u = x + 2\sqrt{y}$

Interior solution, corner solution

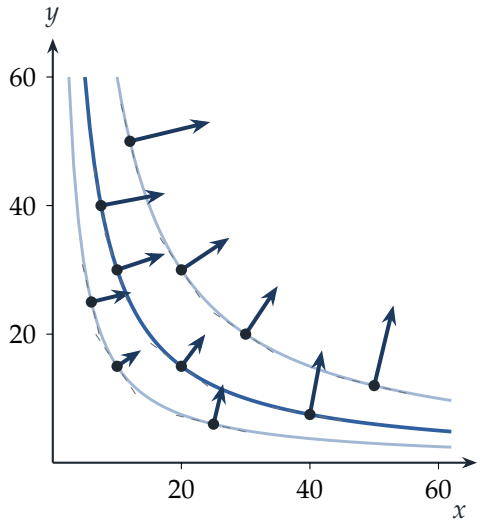
An *interior solution*: some of both goods; the curve just touches the budget line. A *corner solution*: all the money on one good. With Leontief the best bundle is at a *kink*.

- **Linear.** The blue line is $x + y = 60$. Against $3x + y = 60$ the best bundle is on the y axis, $(0, 60)$. Against $x + 3y = 60$ it is on the x axis, $(60, 0)$. Against $x + y = 60$ the budget lies on the curve: every point on the line is best.
- **Leontief.** The kinks lie on $\alpha x = \beta y$. On $3x + y = 60$: $\min\{x, y\}$ has its best bundle at $(15, 15)$, $\min\{3x, y\}$ at $(10, 30)$, $\min\{x, 3y\}$ at $(18, 6)$.

- **Cobb-Douglas.** On $3x + y = 60$: $u = xy$ has its best bundle at $(10, 30)$, $\alpha = 3/4, \beta = 1/4$ at $(15, 15)$, $\alpha = 1/4, \beta = 3/4$ at $(5, 45)$.
- **Quasi-linear.** $u = x + 2\sqrt{y}$ has its best bundle at $(17, 9)$.

4 The gradient

Take many points on various indifference curves. At each point draw the tangent to the curve (dashed), then the gradient (dark arrow).



This is the gradient. We show it with this symbol, ∇u. It is a list of the partial derivatives of the utility function:

$$\nabla u = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}\right).$$

Another way of writing it:

$$\nabla u = (\mathrm{MU}_x, \mathrm{MU}_y).$$

In the picture $u = xy$, so $\nabla u = (y, x)$, drawn at scale $1/4$. It stands at a right angle to the tangent: the tangent has direction $(x, -y)$, and $(y, x) \cdot (x, -y) = 0$.

Where you are on the mountain. Utility is a mountain over the map, and the indifference curves are its height lines. On the slide you drag the point in the 2D graph and see where you are in 3D. The sliders for α and β in $u = x^\alpha y^\beta$ change the curves and the mountain.

5 The budget line and the price arrow

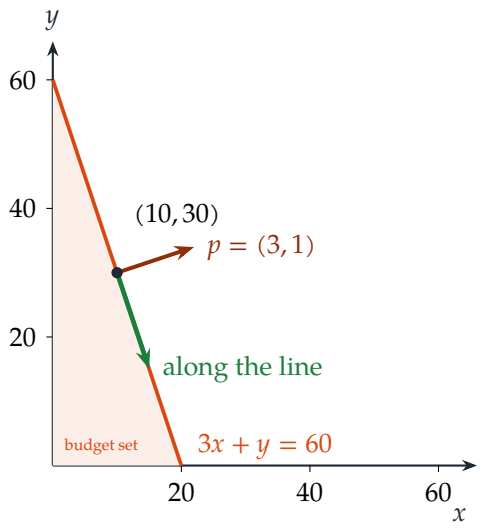
Now the budget comes in, with prices p_x, p_y and money B. From a bundle, one more x costs p_x and one more y costs p_y. Together these make the price arrow

$$p = (p_x, p_y).$$

The arrow along the budget line is $(p_y, -p_x)$. It stands at a right angle to p, because

$$p \cdot (p_y, -p_x) = p_x p_y - p_y p_x = 0.$$

A step along the line costs nothing extra.



The roof. In 3D, spending $p_x x + p_y y$ is a flat roof over the map. The budget set ends at the roof's rim, where spending is B. The wall under the rim stands on the budget line.

6 The mountain and the roof

Now both together. The mountain is utility and the roof is spending. They have different units, so on the slide the roof is scaled so that its rim sits at the best utility u^*. Only the rim and the wall are the constraint. On the wall the mountain reaches the rim at one point only, the best bundle: there the height line at u^* touches the rim. Beyond the wall the mountain goes higher, but you cannot pay for it.

7 Make the two arrows the same

What does it mean for the price vector to match the utility gradient? Assume this function and this cost function:

$$u(x, y) = x\,y, \qquad c(x, y) = p_x x + p_y y = 3x + y.$$

Then $\nabla c = (p_x, p_y)$. One is price, one is utility: we need to convert one to the other. Let's convert price to utility:

$$\nabla u = \lambda \nabla c.$$

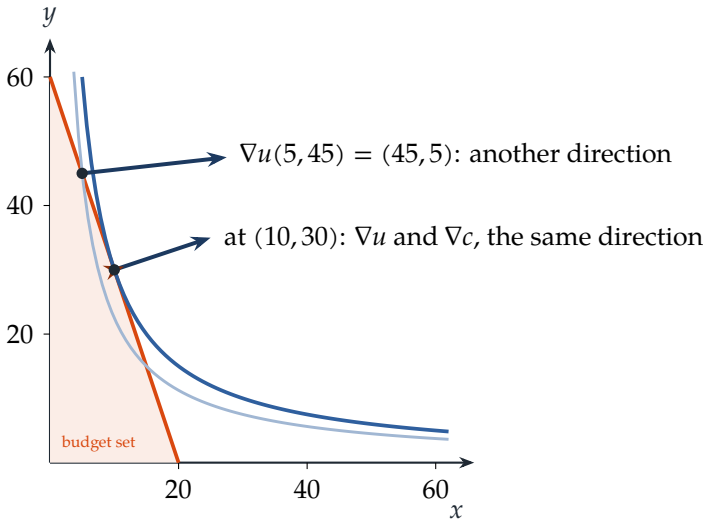
So we match not only the direction, but also the magnitude, so that we can compare them. λ here is the marginal utility of a euro. This vector equals that vector:

$$(\mathrm{MU}_x, \mathrm{MU}_y) = \lambda\,(p_x, p_y),$$

so we need these two equalities:

$$\mathrm{MU}_x = \lambda\, p_x, \qquad \mathrm{MU}_y = \lambda\, p_y.$$

At the best bundle $(10, 30)$: $\nabla u = (30, 10) = 10 \cdot (3, 1)$, so $\lambda = 10$.



The arrows here use one scale of $1/2$.

8 Three pictures, one condition

For $u = xy$ and $c = 3x + y$:

- **Touching curves, the same slope:** $\dfrac{\mathrm{MU}_x}{\mathrm{MU}_y} = \dfrac{p_x}{p_y}$, here $\dfrac{y}{x} = \dfrac{3}{1}$.
- **Bang for the buck:** $\dfrac{\mathrm{MU}_x}{p_x} = \dfrac{\mathrm{MU}_y}{p_y}$, here $\dfrac{y}{3} = \dfrac{x}{1}$.
- **Parallel arrows:** $\nabla u = \lambda\, \nabla c$, here $(y, x) = \lambda\, (3, 1)$.

9 The Lagrangian

Take a utility function of x and y, $u(x, y)$, and the budget $p_x x + p_y y = B$. We set the Lagrangian like this:

$$\mathcal{L} = u(x, y) + \lambda\, (B - p_x x - p_y y).$$

Take the three derivatives and set them to zero:

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial u}{\partial x} - \lambda\, p_x = 0$$

$$\frac{\partial \mathcal{L}}{\partial y} = \frac{\partial u}{\partial y} - \lambda p_y = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = B - p_x x - p_y y = 0$$

We get the following equalities:

$$\mathrm{MU}_x = \lambda p_x, \qquad \mathrm{MU}_y = \lambda p_y, \qquad p_x x + p_y y = B,$$

and we also get the marginal utility of money:

$$\lambda = \frac{\mathrm{MU}_x}{p_x} = \frac{\mathrm{MU}_y}{p_y}.$$

The example. $\mathcal{L} = xy + \lambda(60 - 3x - y)$:

$$\frac{\partial \mathcal{L}}{\partial x} = y - 3\lambda = 0, \qquad \frac{\partial \mathcal{L}}{\partial y} = x - \lambda = 0, \qquad \frac{\partial \mathcal{L}}{\partial \lambda} = 60 - 3x - y = 0.$$

So $x = \lambda$ and $y = 3\lambda$; then $3\lambda + 3\lambda = 60$, so $\lambda = 10$ and $(x, y) = (10, 30)$.

Insight: why the touching point is a best point

It works because the curves bow in. On a curve that bowed out, the touching point would be the *worst* bundle on the line: the two ends of the line would lie on higher curves.

10 Where the derivative is zero

For any function: where its derivative is zero, its change stops. If this is a concave function, that is the highest point of the graph. We can find that maximum by taking the derivative with respect to x and setting it to 0, $f'(x) = 0$, and we are done. On the slide: $f(x) = 10x - x^2$, with $f'(x) = 10 - 2x = 0$ at $x = 5$.

f' is continuous and goes from positive to negative, so by the intermediate value theorem it hits 0 somewhere; concavity makes that point the top. See Osborne §1.4, Proposition 1.4.2 (intermediate value theorem): `https://mjo.osborne.economics.utoronto.ca/index.php/tutorial/index/1/iaf/t#p:ivt`.

Now consider this is the utility function. But wait: our utility function has two variables.

11 A simpler way: one variable

The budget line:

$$3x + y = 60.$$

Define y as a function of x:

$$y = 60 - 3x.$$

The utility function:

$$u = x\,y.$$

Replace y with this equation:

$$u(x) = x\,(60 - 3x) = 60x - 3x^2.$$

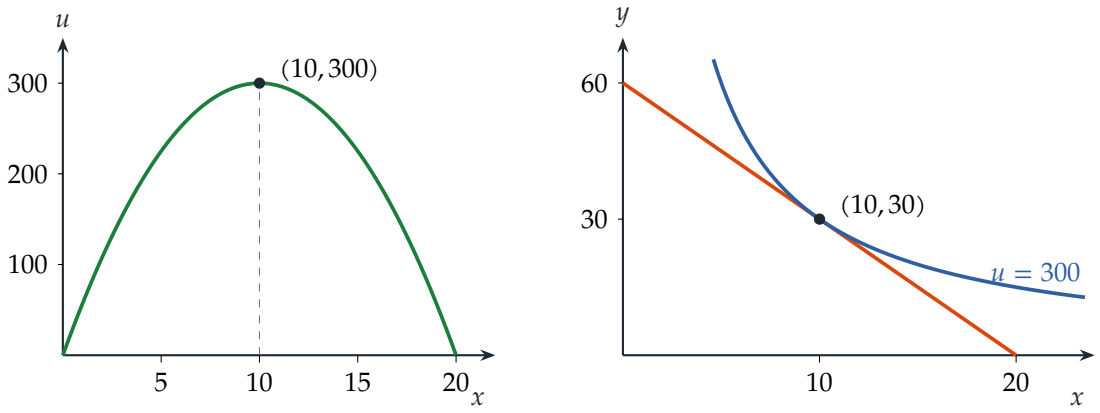
Take the derivative with respect to that one variable:

$$u'(x) = 60 - 6x = 0.$$

So

$$x = 10, \qquad y = 60 - 3 \cdot 10 = 30.$$

The same answer as the Lagrangian. The top of $u(x)$ is the point where the budget line is tangent to the highest indifference curve it reaches.



12 Perfect substitutes

First the simple case again, $u = xy$ on $3x+y = 60$: as you drag the point along the line, $\nabla u = (y, x)$ turns, and $\nabla c = (3, 1)$ stays. At $(10, 30)$ they point the same way.

Now $u = x + y$ on $3x + y = 60$: $\nabla u = (1, 1)$ and $\nabla c = (3, 1)$ stay the same at every bundle. You can never make them match, so the best bundle is the corner $(0, 60)$. Change the price of x: above $p_x = 1$ the best bundle is the corner $(0, 60)$; below $p_x = 1$ it is the corner $(60/p_x, 0)$; at $p_x = 1$ the arrows match and every point on the line is best.

How to solve

1. Draw the budget line.
2. Draw the indifference curves.
3. A corner or a kink? Check the ends and the no-waste line.
4. We have many ways to calculate that point.

We practise them all on the blackboard in the next class, and in Problem Set 4, published Friday evening.

Reading

Book (Frank, *Microeconomics and Behavior*).

- ch. 3, tangency: pp. 69–71
- ch. 3, corner solutions: pp. 71–73
- fixed proportions (Leontief): Concept Check 3.7, p. 73
- App. 3A, marginal utility per euro: pp. 85–87
- App. 3A, Lagrangian: pp. 90–91
- App. 3A, substitution: p. 92

Online (Osborne's math tutorial).

- slopes of level curves (gradient): §2.3, `https://mjo.osborne.economics.utoronto.ca/index.php/tutorial/index/1/13`
- Lagrange: §6.1.1, `https://mjo.osborne.economics.utoronto.ca/index.php/tutorial/index/1/34`
- the multiplier: §6.1.2, `https://mjo.osborne.economics.utoronto.ca/index.php/tutorial/index/1/35`

The Frank pages are in the additional readings of L08 and L09, linked on the week 6 page.